

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-5841
Project Name	FitBuddy – AI Fitness Plan Generator
Maximum Marks	2 Marks

Define Your Technology Stack:

The technologies used for each layer of FitBuddy and the reason each was chosen.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (HTML/CSS/JavaScript)	HTML, CSS, JavaScript with Jinja2 templates	Simple server-rendered pages that load fast, work in any browser and are easy for the team to build and maintain; JavaScript shows the loading overlay.
2	Backend / Server-Side (Python)	Python 3.12, FastAPI, Uvicorn, Pydantic	FastAPI gives quick development, built-in input validation and automatic API docs at /docs; Python matches the team's AI/ML skills.
3	Database / Data Storage (SQL)	SQLite with SQLAlchemy ORM	No setup required for a prototype, stores users and plans in one file, and can be replaced by PostgreSQL by changing DATABASE_URL.
4	Cloud / Hosting / Deployment (Docker)	Docker (python:3.12-slim) and local Uvicorn	Same environment on any machine; the image can be deployed to a cloud service later.
5	Version Control & CI/CD (GitHub)	Git and GitHub; pytest for tests	Team collaboration and change history; automated tests can later be added to a CI pipeline.
6	Third-Party APIs / Other Tools (AI)	Google Gemini API (gemini-2.5-flash) via google-genai	Generates the 7-day workout plan, revises it from feedback and creates the nutrition tip; fast and low cost.